

# Karen Frilya Celine

National University of Singapore | karen.celine@u.nus.edu | karen-celine.github.io

## Education

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**National University of Singapore**, PhD in Computer Science Aug 2021 – present

- Thesis advisors: Warut Suksompong (primary), Frank Stephan, and Sanjay Jain

**National University of Singapore**, BSc (Hons.) in Applied Math Aug 2016 – Aug 2021

- Thesis title: Bi-immunity for Functions
- Thesis advisors: Frank Stephan and Sanjay Jain
- CAP: 4.80/5.0

**National University of Singapore**, BComp (Hons.) in Computer Science Aug 2016 – Aug 2021

- CAP: 4.71/5.0

## Working Paper

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1. Karen Frilya Celine and Warut Suksompong. Optimizing the envy cycle elimination algorithm. *CoRR*, abs/2606.02233, 2026

## Publications

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1. Karen Frilya Celine, Warut Suksompong, and Sheung Man Yuen. On the fairness of additive welfarist rules. *ACM Transactions on Economics and Computation*, Forthcoming. A preliminary version appeared in *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 454–462, May 2025
2. Karen Frilya Celine, Warut Suksompong, and Sheung Man Yuen. Comparing the fairness of recursively balanced picking sequences. In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 69–77, May 2026
3. Karen Frilya Celine, Ziyuan Gao, Sanjay Jain, Ryan Lou, Frank Stephan, and Guohua Wu. Quasi-isometric reductions between infinite strings. *Journal of Computer and System Sciences*, 156:103718, Mar 2026. A preliminary version appeared in *Proceedings of the 49th International Symposium on Mathematical Foundations of Computer Science (MFCS)*, pages 37:1–37:16, Aug 2024
4. Karen Frilya Celine. Balancing fairness and efficiency in the allocation of indivisible goods. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 2920–2922, May 2025. Doctoral Consortium track
5. Karen Frilya Celine, Muhammad Ayaz Dzulfikar, and Ivan Adrian Koswara. Egalitarian price of fairness for indivisible goods. In *Proceedings of the 20th Pacific Rim International Conference on Artificial Intelligence (PRICAI)*, pages 23–28, Nov 2023
6. Cristian S. Calude, Karen Frilya Celine, Ziyuan Gao, Sanjay Jain, Ludwig Staiger, and Frank Stephan. Bi-immunity over different size alphabets. *Theoretical Computer Science*, 894:31–49, Nov 2021

## Thesis

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1. Karen Frilya Celine. Bi-immunity for functions. Bachelor's thesis, National University of Singapore, Singapore, Nov 2020

## Teaching

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**Graduate Tutor**, National University of Singapore Aug 2021 – present

- CS1010S Programming Methodology: 2021/2022 Semester 2
- CS1231S Discrete Structures: 2021/2022 Semester 1, 2022/2023 Semester 2, 2023/2024 Semester 1, 2023/2024 Semester 2, 2024/2025 Semester 1, 2024/2025 Semester 2 (Head TA)
- CS3230 Design and Analysis of Algorithms: 2025/2026 Semester 2 (Head TA)

- CS4261/5461 Algorithmic Mechanism Design: 2024/2025 Semester 1

**Undergraduate Teaching Assistant**, National University of Singapore

Jan – Apr 2021

- CS1231 Discrete Structures: 2020/2021 Sem 2

## Services

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### Reviewer

- Social Choice and Learning Algorithms (SCaLA) 2026
- Games, Agents, and Incentives Workshop (GAIW) 2026

### Subreviewer

- ACM Transactions on Economics and Computation 2026
- Mathematics of Operations Research 2025 – 2026
- International Joint Conference on Artificial Intelligence (IJCAI) 2025

## Internship

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**Software Engineering Intern**, Google Singapore

May – Aug 2020

- Created a microapp to promote Group Buying, by leveraging on Google Pay's Spot Platform.
- Developed the application from scratch in a team of two, including designing the application's user flow, UI/UX mock, database schema and API endpoints, within 14 weeks.
- Responsible for full-stack development of Merchant-related features.

## Awards and Scholarships

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**Research Achievement Award**

Aug 2025

- Awarded for outstanding research performance.

**Full-Time Teaching Assistant Award**

Feb 2025

- Awarded to full-time teaching assistants with exceptional teaching performance.

**ASEAN Undergraduate Scholarship**

Jul 2016 – Jun 2021

- Full scholarship awarded to high-achieving students from ASEAN countries.

**Faculty of Science Dean's List**

Dec 2020

- Awarded to the top 5% of each cohort for academic performance in the preceding semester.

**School of Computing Dean's List**

Dec 2020

- Awarded to the top 5% of each cohort for academic performance in the preceding semester.

**Simon Marais Mathematics Competition Top Decile Individual**

Oct 2020

- Annual undergraduate mathematics competition across Asia Pacific region.

**Dunearn Road Hostel/Sheares Hall Alumni Scholarship**

Sep 2018

- Awarded to residents of Sheares Hall who have done well academically while being actively involved in the hall for the preceding two academic years.

## Skills

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**Programming languages:** Python, JavaScript, Java, C/C++, HTML/CSS, SQL,  $\text{\LaTeX}$ , MATLAB

**Languages:** English (native), Indonesian (native), Malay (proficient), Chinese (basic), Japanese (basic)